Prospective Evaluation of a Point of Care LLM (GTP4) in ICU Outcomes

Protocol Schematic Diagram

CCU Case Identification

- Inclusion and Exclusion Criteria (only medical ER admissions to ICU?)
- 200 cases? (we need to do power calculations)
- ICU Team notifies Research Team Member on call

Acute hypoxic respiratory failure admitted to ICU from ED
<Inclusion Criteria>
* Adults ≥ 18 years
* Diagnosed with AHRF (PaO2/FiO2 ≤ 300 mmHg)
* Requires supplemental oxygen or ventilatory support
<Exclusion Criteria>
* DNI/DNR orders on admission
* Expected death within 48 hours
* Pregnant patients
* Transfer from another hospital with >24h prior treatment

Randomization of Included Cases

- Block Randomization (how do we ideally randomize?)

Standard Group
Traditional resources (No AI)

AI Assisted Group
Traditional Resources + (GTP4)

Physician H&P Clinical Team

Final Assessment and Plan

Physician H&P Clinical Team

Preliminary Assessment and Plan

Clinical Information Extraction Enter into GTP4

Final AI Assisted Assessment and Plan

Clinical Assessment

Focus Questions of Major Interest

- Primary diagnosis
- Differential diagnosis (5 most likely diagnoses)
- Therapeutic Plan (5 key points)

Clinical Team (Resident and Attending)

Research Team (on call member)

- Remove Assessment and Plan
- Deidentify (double screen for patient confidentiality)
- Fill Template (we need to refine template)
- GTP4 Input/Output

Clinical Team (Resident and Attending)

- Clinical Team is provided with GTP4 data
- Reanalysis of Assessment and Plan
- Questionnaire: did AI change anything?

the AI tool will provide :
* 5 differentials and plans
* Recommendations on intubation and/or ventilator settings
* Risk predictions for clinical deterioration
* Alerts requiring escalation of care

Data Collection

Confidential Information
To remain in MetroWest Medical Center Computer System

- Double screened for patient confidentiality
- Only redacted H&P is removed from Tenet system

Subgroup analysis
(1) etiologise : pneumonia, Heart failure with pulmonary edema, ILD flare, COPD with acute exacerbation etcs
(2) age (cutoff : 65 year old ?)
(3) BMI > 30 (OSA and obesity hypoventilation syndrome)
(4) Smoking : 20 pack per year
(5) severity of hypoxia : PF ratio 200-300, 100-200, <100
(6) with or without underlying disease : cardiac history (coronary artery disease or not, heart failure), DM, COPD, ILD. chronic kidney disease

Analysis

Standard Group
Longitudinal Assessment
"Ground Truth"
- Outcomes

Compare

AI Assisted Group
Longitudinal Assessment
"Ground Truth"
- Outcomes

Primary outcome :
(1) In-hospital all-cause mortality
Secondary outcome :
(1) ICU and hospital length of stay
(2) Ventilator-free days at day 28
(3) Time to escalation (e.g., intubation)
(4) Incidence of complications (e.g., VAP and ARDS)
(5) Abx/steroid duration

- Outcomes (what clinical outcomes do we want to compare?
- Primary vs Secondary Outcomes
- Group Comparison
- Subgroup Analysis (refer to the left side)
- Statistical Methods

Statistical methods:
* Chi-square or logistic regression for primary outcome
* t-test or Mann-Whitney U for continuous variables
* Cox proportional hazards for time-to-event analyses
Intention-to-treat (ITT) and per-protocol analyses will be performed